WORKSHEET

Moving means

1 For each of the following tables of time series data:

i complete the table by calculating the three-point moving average, rounded to one decimal place where appropriate

ii use CAS or a spreadsheet to plot both the original and the smoothed data.

a Unemployment in the town of Bargon, 2012

Month	No. of unemployed	Three-point moving average
January	678	
February	645	
March	672	
April	641	
May	628	
June	638	
July	615	
August	618	
September	593	
October	581	
November	599	
December	573	

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b Number of bands performing at a music festival in Sydney from 1992 until 2011

Year	No. of bands performing	Three-point moving average
1992	21	
1993	17	
1994	25	
1995	31	
1996	44	
1997	50	
1998	41	
1999	37	
2000	37	
2001	38	
2002	35	
2003	40	
2004	49	
2005	48	
2006	48	
2007	55	
2008	41	
2009	44	
2010	66	
2011	48	

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c Number of low-birthweight babies born in a suburban hospital from 1997 to 2013

Year	No. of low-birthweight babies born	Three-point moving average
1997	65	
1998	81	
1999	63	
2000	56	
2001	62	
2002	57	
2003	45	
2004	70	
2005	48	
2006	55	
2007	38	
2008	67	
2009	71	
2010	59	
2011	80	
2012	49	
2013	53	

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2 For each of the following tables of time series data:

i complete the table by calculating the two- and four-point moving average with centring to smooth the data. Round to one decimal place where appropriate.

ii Use CAS or a spreadsheet to plot both the original and the smoothed data.

a Number of houses sold each year by a real-estate business

Year	No. of houses sold	Four-point moving average	Four-point moving average with centring
2000	28		
2001	35		
2002	41		
2003	21		
2004	29		
2005	21		
2006	16		
2007	32		
2008	11		
2009	19		
2010	14		

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2011	29		
2012	21		
2013	33		

b Population of the town of Wagga Wagga

Year	Pop'n	Four-point moving average	Four-point-moving average with centring
2001	51 714		
2002	52 148		
2003	52 356		
2004	52 673		
2005	53 195		
2006	53 901		
2007	54 812		
2008	55 582		
2009	56 185		
2010	56 566		
2011	56 788		

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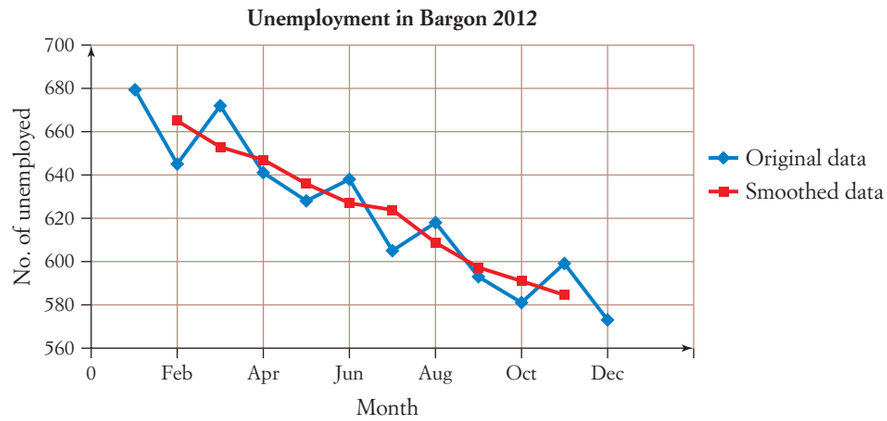
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Answers

1 a i

Month	No. of unemployed	Three-point moving average
January	678	
February	645	$\frac{678 + 645 + 672}{3} = 665$
March	672	$\frac{645 + 672 + 641}{3} = 652.7$
April	641	$\frac{672 + 641 + 628}{3} = 647$
May	628	$\frac{641 + 628 + 638}{3} = 635.7$
June	638	$\frac{628 + 638 + 615}{3} = 627$
July	615	$\frac{638 + 615 + 618}{3} = 623.7$
August	618	$\frac{615 + 618 + 593}{3} = 608.7$
September	593	$\frac{618 + 593 + 581}{3} = 597.3$
October	581	$\frac{593 + 581 + 599}{3} = 591$
November	599	$\frac{581 + 599 + 573}{3} = 584.3$
December	573	

ii



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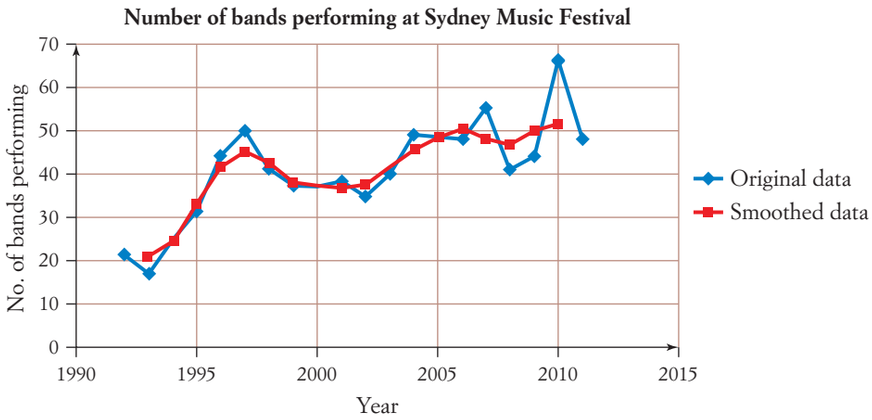
b i

Year	No. of bands performing	Three-point moving average
1992	21	
1993	17	$\frac{21 + 17 + 25}{3} = 21$
1994	25	$\frac{17 + 25 + 31}{3} = 24.3$
1995	31	$\frac{25 + 31 + 44}{3} = 33.3$
1996	44	$\frac{31 + 44 + 50}{3} = 41.7$
1997	50	$\frac{44 + 50 + 41}{3} = 45$
1998	41	$\frac{50 + 41 + 37}{3} = 42.7$
1999	37	$\frac{41 + 37 + 37}{3} = 38.3$
2000	37	$\frac{37 + 37 + 38}{3} = 37.3$
2001	38	$\frac{37 + 38 + 35}{3} = 36.7$
2002	35	$\frac{38 + 35 + 40}{3} = 37.7$
2003	40	$\frac{35 + 40 + 49}{3} = 41.3$
2004	49	$\frac{40 + 49 + 48}{3} = 45.7$
2005	48	$\frac{49 + 48 + 48}{3} = 48.3$
2006	48	$\frac{48 + 48 + 55}{3} = 50.3$
2007	55	$\frac{48 + 55 + 41}{3} = 48$
2008	41	$\frac{55 + 41 + 44}{3} = 46.7$
2009	44	$\frac{41 + 44 + 66}{3} = 50.3$
2010	66	$\frac{44 + 66 + 48}{3} = 52.7$
2011	48	

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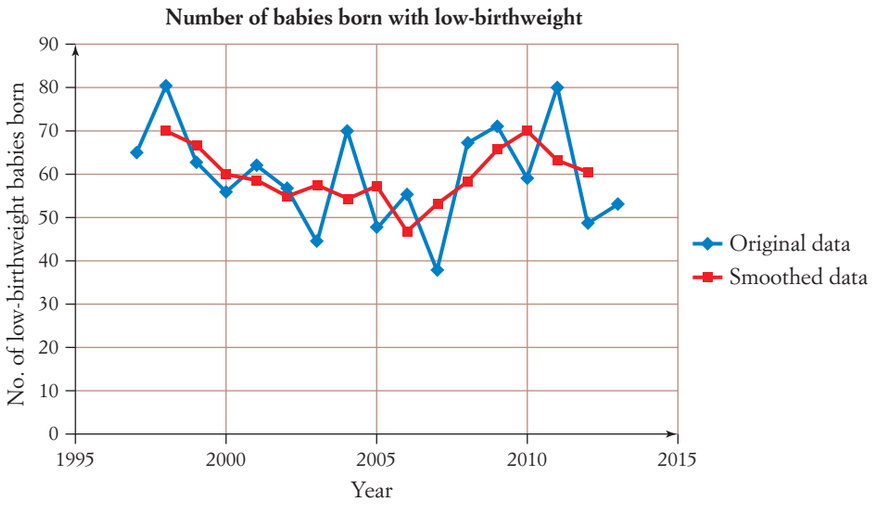
c i

Year	No. of low-birthweight babies born	Three-point moving average
1997	65	
1998	81	$\frac{65 + 81 + 63}{3} = 69.7$
1999	63	$\frac{81 + 63 + 56}{3} = 66.7$
2000	56	$\frac{63 + 56 + 62}{3} = 60.3$
2001	62	$\frac{56 + 62 + 57}{3} = 58.3$
2002	57	$\frac{62 + 57 + 45}{3} = 54.7$
2003	45	$\frac{57 + 45 + 70}{3} = 57.3$
2004	70	$\frac{45 + 70 + 48}{3} = 54.3$
2005	48	$\frac{70 + 48 + 55}{3} = 57.7$
2006	55	$\frac{48 + 55 + 38}{3} = 47$
2007	38	$\frac{55 + 38 + 67}{3} = 53.3$
2008	67	$\frac{38 + 67 + 71}{3} = 58.7$
2009	71	$\frac{67 + 71 + 59}{3} = 65.7$
2010	59	$\frac{71 + 59 + 80}{3} = 70$
2011	80	$\frac{59 + 80 + 49}{3} = 62.7$
2012	49	$\frac{80 + 49 + 53}{3} = 60.7$
2013	53	

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2 a i

Year	No. of houses sold	Four-point moving average	Four-point moving average with centring
2000	28		
2001	35		
		$\frac{28 + 35 + 41 + 21}{4} = 31.3$	
2002	41		$\frac{31.3 + 31.5}{2} = 31.4$
		$\frac{35 + 41 + 21 + 29}{4} = 31.5$	
2003	21		$\frac{31.5 + 28}{2} = 29.8$
		$\frac{41 + 21 + 29 + 21}{4} = 28$	
2004	29		$\frac{28 + 21.8}{2} = 24.9$
		$\frac{21 + 29 + 21 + 16}{4} = 21.8$	
2005	21		$\frac{21.8 + 24.5}{2} = 23.2$
		$\frac{29 + 21 + 16 + 32}{4} = 24.5$	
2006	16		$\frac{24.5 + 20}{2} = 22.3$
		$\frac{21 + 16 + 32 + 11}{4} = 20$	

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2007	32		$\frac{20 + 19.5}{2} = 19.8$
		$\frac{16 + 32 + 11 + 19}{4} = 19.5$	
2008	11		$\frac{19.5 + 19}{2} = 19.3$
		$\frac{32 + 11 + 19 + 14}{4} = 19$	
2009	19		$\frac{19 + 18.3}{2} = 18.7$
		$\frac{11 + 19 + 14 + 29}{4} = 18.3$	
2010	14		$\frac{18.3 + 20.8}{2} = 19.6$
		$\frac{19 + 14 + 29 + 21}{4} = 20.8$	
2011	29		$\frac{20.8 + 24.3}{2} = 22.6$
		$\frac{14 + 29 + 21 + 33}{4} = 24.3$	
2012	21		
2013	33		

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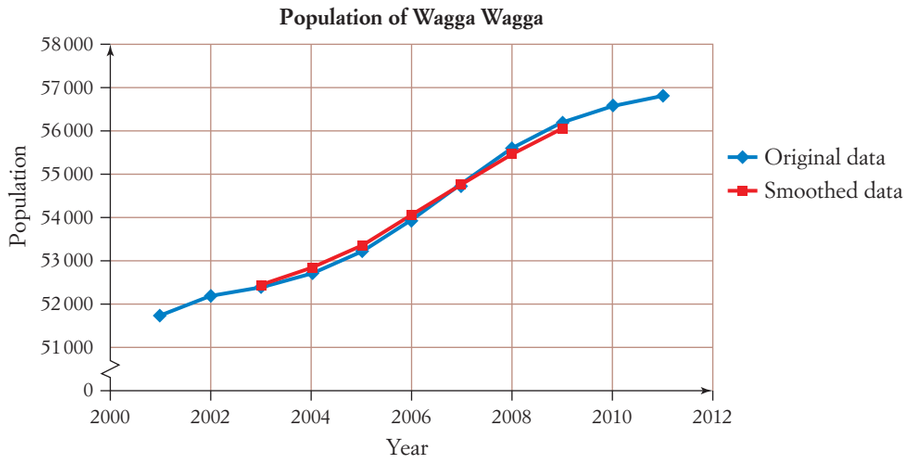
b i

Year	Pop'n	Four-point moving average	Four-point moving average with centring
2001	51 714		
2002	52 148		
		$\frac{51714 + 52148 + 52356 + 52673}{4} = 52\,222.8$	
2003	52 356		$\frac{52222.8 + 52593}{2} = 52\,407.9$
		$\frac{52148 + 52356 + 52673 + 53195}{4} = 52\,593$	
2004	52 673		$\frac{52593 + 53031.3}{2} = 52\,812.2$
		$\frac{52356 + 52673 + 53195 + 53901}{4} = 53\,031.3$	
2005	53 195		$\frac{53031.3 + 53645.3}{2} = 53\,338.3$
		$\frac{52673 + 53195 + 53901 + 54812}{4} = 53\,645.3$	
2006	53 901		$\frac{53645.3 + 54372.5}{2} = 54\,008.9$
		$\frac{53195 + 53901 + 54812 + 55582}{4} = 54\,372.5$	
2007	54 812		$\frac{54372.5 + 55120}{2} = 54\,746.3$
		$\frac{53901 + 54812 + 55582 + 56185}{4} = 55\,120$	
2008	55 582		$\frac{55120 + 55786.3}{2} = 55\,453.2$
		$\frac{54812 + 55582 + 56185 + 56566}{4} = 55\,786.3$	
2009	56 185		$\frac{55786.3 + 56280.3}{2} = 56\,033.3$
		$\frac{55582 + 56185 + 56566 + 56788}{4} = 56\,280.3$	
2010	56 566		
2011	56 788		

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